World Health Organization

Syndromic screening for infection prevention and control measures during public health emergencies

Technical note

Key messages

- **Syndromic screening is a systematic approach to identifying persons with signs and symptoms of infection at earliest opportunity in health-care settings, enabling timely identification and implementation of enhanced infection prevention and control (IPC) measures.**
- **Systematically screen all patients – both inpatients and outpatients – for signs and symptoms of suspected infectious rash, infectious diarrhoea or acute respiratory infection.**
- **In addition to standard precautions, apply transmission-based precautions relevant to suspected high-threat pathogens when identified by syndromic screening.**

Intended audience

This technical note is intended for hospital administrators, emergency unit managers, primary care managers, emergency management focal persons, field epidemiologists, health and care workers and IPC professionals.

Objective

The objective of this technical note is to inform health-care settings of a simplified screening approach for early detection and safe placement of patients with syndromes of suspected infections that are known to rapidly transmit as health care-associated infections. This technical note includes contextual information relevant to implementing syndromic screening for IPC measures in the context of public health emergencies.

This technical note contains three parts, which may be referred to based on individual health systems' needs or as a combined approach to enable early detection; put in place administrative and engineering IPC measures; and establish a link between syndromic screening and HAI surveillance and/or syndromic surveillance systems. An annex section is included with example tools to support passive and active syndromic screening for IPC.

I. Syndromic screening for epidemiologically significant pathogens
II. Administrative and engineering controls for screening and patient placement
III. Linking syndromic screening with HAI surveillance and syndromic surveillance systems

Background

Syndromic screening has been applied in a variety of health-care and community settings to collect syndromic surveillance data on individual and population health indicators *(1-3)*. Aggregate syndromic surveillance data provides a point-in-time epidemiological indicator, which may trigger additional actions after it has been reported through early warning, alert and response (EWAR) and similar mechanisms (*2-5)*. In addition to collecting epidemiological data for syndromic surveillance, syndromic screening has been utilized in various settings as an applied form of risk assessment for the immediate implementation of IPC measures *(2, 6-10).* Key concepts relevant to syndromic screening for IPC are defined in *Table 1* below.

Table 1. Key concepts relevant to syndromic screening for IPC during public health emergencies

Acuity-based triage	Acuity-based triage is the action of sorting and prioritizing patients based on the estimation of the urgency for clinical intervention. Triage is used to identify which patients require immediate medical intervention and which ones can safely wait. Acuity-based triage can be performed at any point of access to the health-care system, including in both pre-hospital ambulance services and hospitals *(11)*.
Early warning, alert and response (EWAR)	The organized mechanism to rapidly detect and respond to signals that might indicate potential acute public health events *(2).*
Event-based surveillance	The organized collection, monitoring, assessment and interpretation of mainly unstructured, ad-hoc information regarding potential public health events or risks that may represent an acute risk to human health *(2, 3)*.
Health care-associated infection (HAI)	An infection occurring in a patient during the process of care in a hospital or other health care facility, which was not present or incubating at the time of admission. Health care-associated infections can also appear after discharge. They represent the most frequent adverse event associated with patient care *(12)*.
Incident-based surveillance	The systematic collection, monitoring, analysis and interpretation of structured, health-related data (indicators) produced by health facilities or other defined sources. Reporting is based on standardized case definitions of selected priority diseases or conditions *(2)*.
Risk assessment	An IPC standard precaution involving continuous scanning of the health-care environment and selection of the appropriate actions and equipment needed to reduce the risk of exposure by patients or health and care workers to infectious agents. This includes identifying patients with symptoms of undiagnosed infection(s) and managing this risk by wearing risk-appropriate personal protective equipment (PPE), spatially separating patients who pose a risk of transmission to others and initiating transmission-based precautions *(derived from (13))*.
Standard precautions	The minimum standard of IPC practices that should be used by all health and care workers during the care of all patients, at all times, in all health-care settings *(13)*.
Syndromic screening	The process involved in collecting syndromic surveillance information directly from individual persons using one or more methods, such as temperature check, observation and health declaration forms, with subsequent follow-up interventions (including testing and other health measures) *(derived from (1, 3, 10))*.
Syndromic surveillance	The detection of individual and population indicators of illness that are discernible before confirmed laboratory diagnoses are made *(3)*.
Transmission-based precautions	Specific IPC precautions used in addition to standard precautions for patients with known or suspected infection or colonization with epidemiologically significant pathogens that can be transmitted during the delivery of health care (e.g. contact, droplet, airborne) *(14)*.

I. Syndromic screening for epidemiologically significant pathogens

Syndromic screening is intended to be performed both at first point of contact, where patients present to the health-care setting, and as part of standard, daily clinical assessment of inpatients *(8, 12)*. The timely identification, spatial separation, and application of transmission-based precautions to patients with suspected infections that can transmit in health-care settings is intended to reduce the risk of infectious disease transmission *(13)*.

A syndromic screening approach may also be used to screen health and care workers for symptoms of infection as they arrive at health facilities. However, other mechanisms for self-reporting of staff sickness

should be integrated locally with occupational health and safety programmes. Once a health or care worker has been identified as sick, follow-up measures require actioning steps beyond transmission-based precautions, such as exclusion from work criteria and occupational exposure investigations in cases where the syndrome is suspected to have resulted from an exposure within the health-care setting *(15)*.

Passive syndromic screening

Passive syndromic screening performed in health-care settings relies on patients and visitors to self-identify signs and symptoms (syndromes) of epidemiologically significant pathogens. This may be initiated using a poster or banner (i.e. in the waiting room or at the entrance), video or other prompt to engage patients and visitors to self-screen, to take actions to prevent infections in others and to inform screening staff (*see example poster in Annex 2*) *(8, 16)*. Passive syndromic screening may increase the timeliness of identification when accompanied by clear direction about the expectation of actions to be taken by the patient or visitor (e.g. hand hygiene, use of medical mask, self-reporting of potential infection) *(8, 9)*.

Active syndromic screening

Active screening entails proactively monitoring for signs or symptoms by engaging with patients through questionnaires, electronic formats, or verbal inquiries *(16)*. It involves directly assessing patients' current status, such as conducting temperature checks, and asking patient to self-identify apparent symptoms of suspected infection (*see example screening tool in Annex 3*) *(7, 10)*. In addition, active syndromic screening may be used to identify patients with compatible symptoms + recent travel history or contact with a sick traveller, indicating potential exposure to a high-threat pathogen *(1)*.

Local outbreaks and high-threat pathogens

Additional syndromic screening criteria that are locally relevant to endemic, epidemic and emerging pathogens may be considered using standardized criteria (e.g. community case definitions). In the instance of a local or national-level outbreak of a high-threat pathogen requiring immediate patient placement and application of risk-appropriate, transmission-based precautions (e.g. precautions specific to viral haemorrhagic fevers) *(17, 18)*, specific screening question(s) and follow-up actions should be added to syndromic screening for suspected high-threat pathogens with unique symptom presentations.

To aid in the development and maintenance of local or nationally applicable screening criteria for health and care workers performing screening for travel-related, high-threat pathogens, a review of continuously updated resources at global level can be performed[1] *(19-21)*. National or regional public health authorities may also include updated alert mechanisms for ongoing outbreaks and high-threat pathogens of interest and should be referenced to incorporate in locally applicable syndromic screening practices *(22, 23)*.

Syndromic screening criteria and actions for IPC

At all entry points to health-care settings (e.g. emergency units, outpatient clinics, hospital admissions, etc.) and as part of ongoing risk assessment when caring for patients admitted to the health facility, ask about or assess patients for signs and symptoms of potential infection:

- Does patient have an unexplained rash or lesions on their face or body? If yes:
    - If rash or lesions are present on face or have disseminated across body:
        - Provide patient with a medical mask and directions on how to wear it.
        - Ask or assist patient to wash hands.

[1] Examples of continuously updated resources providing alert of outbreaks of high-threat pathogens include: World Health Organization's Disease Outbreak News *(19)*, International Society for Infectious Diseases Programme for Monitoring Emerging Diseases (ProMED) *(20)*, Healthmap.org *(21)*.

        - Immediately place patient in a private room (an airborne infection isolation room, where available) on airborne/contact precautions until further clinical investigation can occur *(13)*.
    - If rash or lesions are localized/non-disseminated on body and fully covered:
        - Provide patient with a medical mask and directions on how to wear it.
        - Ask or assist patient to wash hands.
        - Allow patient to remain in waiting room per acuity-based triage priority if rash or lesions are localized and fully covered by clothing or bandaging.
        - When patient can be seen per acuity-based triage priority *(11)*, place patient in a private room and apply contact precautions until further clinical investigation can occur *(13)*.

- Does patient have new or worsening diarrhoea (passing at least three loose or watery stools in the previous 24 hours or more than baseline if patient regularly experiences loose or watery stool)?
    - If yes:
        - Ask or assist patient to perform hand hygiene.
        - Ask about or assess risk of incontinence of stool; if likely, immediately provide dedicated patient space, including toileting/latrine *(24)*.
        - Allow patient to remain in waiting room per acuity-based triage priority if there is no immediate risk of incontinence *(24)*.
        - When patient can be seen per acuity-based triage priority *(11)*, place patient in a private room on contact precautions with dedicated toileting until further clinical investigation can occur *(13)*.

- Has patient had respiratory symptoms (new or worsening cough, runny nose/sneezing, or shortness of breath and fever or chills in the previous 24 hours)?
    - If yes:
        - Provide patient with a medical mask and directions on how to wear it *(9).*
        - Ask or assist patient to wash hands.
        - Allow patient to remain in waiting room per acuity-based triage priority if medical mask is tolerated/worn correctly and occupancy of waiting room does not exceed availability of space or adequate ventilation rate *(25-27)*.
        - When patient can be seen per acuity-based triage priority *(11)*, place patient in a private room with adequate ventilation and on droplet precautions until further clinical investigation can occur *(6)*.

- Does patient have any of the above symptoms and has the patient recently travelled to an area with an outbreak (epidemic) or endemic risk of transmission of a high-threat pathogen or interacted with a sick person who has recently travelled to such an area?
    - If travel history is not relevant:
        - Apply standard and transmission-based precautions as described above or as indicated by risk assessment *(13, 14)*.
    - If travel history to an area with a known, high-threat pathogen is relevant:
        - Review screening criteria (if available) or other resources to determine if the symptoms the patient is presenting with are compatible with a high-threat pathogen of interest in the area of the patient's recent travel or sick traveller contact.
        - If symptoms compatible with travel alerts are identified, apply transmission-based precautions relevant to the suspected high-threat pathogen and place the patient in a private room until further clinical investigation can occur.

Clinical investigation for patients placed on transmission-based precautions

After conducting syndromic screening and initiating appropriate placement of patients with suspected infections, perform further clinical assessment to develop a preliminary diagnosis of the suspected infectious or non-infectious cause(s) of symptoms. Rapid diagnostic testing and other forms of microbiological testing should also be considered during the clinical investigation step. The clinical investigation step may identify a need to change the type of transmission-based precautions needed for safe patient care or may indicate that care may be performed using standard precautions alone, depending on the suspected diagnosis *(13)*.

Examples of specific syndromes indicating suspected infections that can be identified through syndromic screening may be found in WHO's *Transmission-based precautions for the prevention and control of infections: aide-memoire (13)*.

II. Administrative and engineering controls for screening and patient placement

At first point of access to the health-care facility/screening areas:

- Ensure a functional hand hygiene station equipped with alcohol-based handrub or soap and clean water is located at or before the entrance to the health-care facility *(derived from (14))*.
- Display hand hygiene signage that encourages all patients, visitors and staff members to perform hand hygiene when entering the health-care facility *(derived from (14))*.
- Ensure waste bins and bags are available.
- Display posters or banners for passive syndromic screening for IPC (see Annexes) to enable patients and visitors to self-screen and take appropriate actions to mitigate transmission to others.
- Ensure the availability of medical masks meeting minimum technical specifications for use by patients with suspected infectious rash or respiratory infections *(29)*.
- Ensure PPE that meets or exceeds WHO's recommended technical specifications (e.g. medical masks, filtering facepiece respirators, eye protection, fluid-resistant gowns, gloves) are stocked and accessible for health and care worker use *(29, 30)*.
- Ensure patients are seated at least 1 metre apart in waiting areas *(26)*.
- Ensure good indoor ventilation in waiting areas with a minimum ventilation rate of 2.5 litres per second per cubic metre of space volume[2] *(25)*.
- During active screening, ensure that health and care workers are at least 1 metre from patients *(14, 26)*.
    - Use digital (no-touch) thermometers for patient temperature (fever) assessment *(26)*.

Patient placement

Ensure appropriate placement of patients with suspected infections.

- Use signage at the doorway of the patient room to indicate when a patient is placed on transmission-based precautions (e.g. contact, droplet, airborne) *(13)*.
- Patients requiring transmission-based precautions during care should be placed in well-ventilated single rooms, where feasible *(26)*.
    - In general, all inpatient rooms are recommended to have a ventilation rate of at least 60 litres per second per patient *(25)*.
    - For airborne infection isolation rooms, it is recommended to have a ventilation rate of 160 litres per second *(25, 26)*.
        - If high-efficiency portable air (HEPA) filtration units are used *(26)*, they should meet minimum efficiency reporting value (MERV) 14 (ISO ePM1) 70-80% filters or higher and be maintained according to manufacturer specifications *(27)*.

[2] The World Health Organization has developed an online calculator to assist estimation of risk of airborne transmission in indoor congregate settings: *Airborne Risk Indoor Assessment (ARIA)* *(28)*.

- Patients placed on contact precautions should be provided with dedicated toileting/latrine *(13, 24)*.
    - If dedicated toilet/latrine is unavailable, ensure the availability of a bedpan or commode and handwashing station as an alternative.
- Minimize unnecessary movement and transportation of patients outside their rooms or designated areas.
    - If patient transport becomes necessary, use predetermined transport routes designed to minimize exposure to health and care workers, other patients and visitors *(derived from (13))*.
    - For a patient with suspected or confirmed infectious rash or acute respiratory infection, provide the patient with a medical mask to wear, if tolerated, during the transport *(13)*.
        - Ensure exposed rashes are covered during patient transport *(derived from (13))*.

Special consideration – overcrowding and cohorting

Cohorting is the act of grouping patients who are colonized or infected with the same organism with the aim to confine their care to one area and prevent them from coming into contact with other susceptible patients. Cohorts are created based on clinical diagnosis, microbiological confirmation with available, epidemiology and the mode of transmission of the infectious agent. Cohorting is reserved for situations where there is an insufficient number of single rooms and where patients have laboratory confirmed infections with the same pathogen which can be grouped together – away from other patients. Dedicated equipment, toilets and staff should be used for patients within the area reserved for the cohort.

In a situation where single rooms are not available, or where the bed occupancy rate is expected to reach 100% or more (overcrowding), it remains not advisable to cohort non-laboratory confirmed infections due to the risk of cross-transmission of different types of infections *(derived from (24))*.

When health-care facility occupancy rates are expected to be exceeded, consider the following options:

- Assess existing space within the health-care facility to determine if it can be modified to make available additional space for patient care *(31)*.
- Establish safe referral and transfer systems to partnered health-care facilities *(31)*.
- During intense surges in patient care needs or during community outbreaks of infectious diseases, consider establishing or repurposing existing community infrastructure temporarily to function as surge capacity *(31)*.

General considerations for the use of cohorting are provided below. However, these should always be applied in conjunction with specific advice from IPC guidance for the suspected or confirmed infection of interest.

Laboratory-confirmed cases who are identified with the same infection AND are not suspected or confirmed to carry any other infection that can be transmitted in health-care settings, may be considered for cohorting *(derived from (24))*.

Non-laboratory-confirmed cases of the same suspected infection may be placed in cohort-isolation if symptoms and diagnostic suspicion are compatible with an infection that is either endemic or in known outbreak, as indicated by local epidemiology (e.g. *Vibrio cholerae*). In instances where multiple infections causing similar symptoms are known to be transmitting locally, and where laboratory confirmation cannot be enabled, cohorting should be avoided *(derived from (24))*.

III. Linking syndromic screening with HAI surveillance and syndromic surveillance systems

Active syndromic screening of inpatients for IPC measures and HAI surveillance

Health and care workers, including IPC professionals, should continue to apply active syndromic screening when monitoring inpatients for new onset of symptoms that were not present during the time of admission *(32)*. Immediately implement transmission-based precautions for patients who develop suspected infections of concern during the course of inpatient care and evaluate the possibility of HAI *(12, 13)*.

Additional IPC actions, reporting to relevant public health authorities, and confirmatory diagnostic testing are recommended in instances where active syndromic screening or syndromic surveillance systems identify multiple patients experiencing onset of similar symptoms after admission to a health facility, as this may indicate a cluster or outbreak of HAIs *(derived from (32))*.

Interface with syndromic surveillance systems

During public health emergencies, incident- and event-based reporting mechanisms such as EWAR or nationally notifiable disease surveillance systems may be in place to collect incident-based reporting of cases *(2)*. Incident-based reporting can be performed as a complementary activity alongside syndromic screening for IPC. With some adaptation, syndromic surveillance systems can be enhanced with criteria to identify and report alerts of suspected incidents of HAIs (e.g. if date of symptom onset is >48 hours after date of admission or meets criteria of an HAI case definition for selected infections of interest) *(12)*. Event-based reporting should additionally be considered where clusters of patients are identified to have developed symptoms after admission to the health facility.

Some considerations for the integration of alerts for suspected HAIs into syndromic surveillance systems (and their equivalent) during public health emergencies include:

- Syndromic surveillance data collection processes require specific information to be able to report incident-based surveillance when a current inpatient develops new onset of symptoms meeting an HAI or community case definition.
    - "Date(s) of admission" and "Date of symptom onset" (as a minimum) should be included as part of standardized case investigation forms or other data collection processes for current inpatients or readmissions to enable generation of alerts for suspect HAI *(12)*.
- Syndromic surveillance data collection processes would additionally benefit from pre-established criteria and/or forms to identify and report event-based surveillance priorities for suspected clusters and outbreaks occurring in health-care facilities *(2)*.
    - An example of an event of interest may include an unusual increase in the number of patients linked by a similar exposure to the same room, ward or procedure and identified with compatible symptoms of infection after admission to a health facility *(12)*.

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Annex 1. Methods for development

This technical note was developed through consolidation of existing WHO documents, such as guidelines, guidance documents, fact sheets and training modules. Such documents were selected from the WHO database based on their applicability and importance to and impact on IPC practice. External resources on syndromic screening and syndromic surveillance; including policy documents endorsed by national and sub-national authorities, were provided by external experts during the review phase and were selected by WHO for inclusion in the technical note based on their established reputation, practical utility and regular application in practice.

The IPC Public Health Emergencies Working Group reviewed the draft document using Microsoft SharePoint by email request, as did various other units within WHO, including focal points within the six WHO regions; their feedback was integrated into the final document. Declarations of interests have been collected and assessed for all external reviewers. Due diligence and risk assessment in accordance with the provisions of the Framework for Engagement with Non-State Actors (FENSA) has been conducted on entities that are part of the Global Infection Prevention and Control Network, including the IPC in Public Health Emergencies Working Group. Conflict of interest declaration forms for external reviewers were collected and reviewed according to WHO policy. No conflicts of interest were identified. For more information on authorship and contributions, please refer to the Acknowledgements section.

World Health Organization

Attention patients and visitors

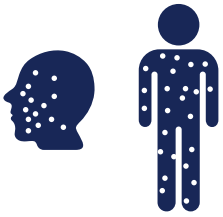
Always wash your hands when you are entering the health-care facility

Please inform staff if you have:

1

Unexplained **rash or lesions** on your face or body.

If yes, wash your hands, wear a mask and inform clinic staff right away.



2

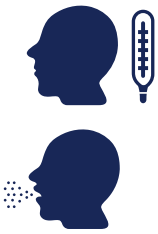
New or worsening **diarrhoea** in the past 24 hours.

If yes, wash your hands and inform clinic staff if you have immediate need for toileting.

3

New or worsening **cough**, **shortness of breath**, or **runny nose/sneezing** AND **fever** or **chills** in the past 24 hours.

If yes, wash your hands and wear a mask or request a mask from clinic staff.



4

Answered "yes" to any of the above AND you have **travelled** in the last 21 days, OR you have been with a **sick person who has travelled** in the last 21 days.

If yes, wash your hands and inform clinic staff right away.

Active Syndromic Screening for Infection Prevention and Control

World Health Organization

For use by **staff** to assess patients

Ask or assess patient:	If yes:
1 Do you have an unexplained visible **rash or lesions** on your face or body?	• Provide patient with medical mask and directions on how to use it • Ask patient to perform hand hygiene • If rash or lesions are localized and covered by clothing, apply contact precautions and place in single patient room with dedicated toileting for further assessment • If rash or lesions are present on face or full body, apply contact and airborne precautions and place patient in well-ventilated single room (preferably airborne infection isolation room) for further assessment
2 Do you have new or worsening **diarrhoea** in the past 24 hours?	• Ask patient to perform hand hygiene • Apply contact precautions • Place patient in single room with dedicated toileting for further assessment
3 Do you have new or worsening **cough**, **shortness of breath**, or **runny nose/sneezing** AND **fever** or **chills** in the past 24 hours?	• Provide patient with medical mask and directions on how to use it • Ask patient to perform hand hygiene • Apply droplet precautions and use eye protection • Place patient in well-ventilated single room for further assessment
4 Did you answer "yes" to any of these questions AND • You have **travelled** in the last 21 days OR • You have been in direct contact with a sick person who has travelled within the last 21 days?	• Apply standard and transmission-based precautions with patient placement as described above • Review desk reference (if available) of known locations with cases or outbreaks of high-threat pathogens • Escalate transmission-based precautions or flag for clinical assessment to consider escalation if symptoms and travel/contact history are compatible with high-threat pathogen risk

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